

Attention Enhanced Multi-Agent Reinforcement Learning for Cooperative Spectrum Sensing in Cognitive Radio Networks

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Abstract—Cooperative spectrum sensing (CSS) technology has been widely studied to enhance the spectrum sharing efficiency spatially and temporally in cognitive radio networks (CRNs), where the secondary users (SUs) can opportunistically reuse the channels already licensed to the primary users (PUs) for transmission by sensing spectrum holes. SUs are endowed with the global awareness of channels state by cooperating with each other without sweeping across the whole frequency bands. Since the channels occupation of PUs changes dynamically, the accurate sensing and swift information sharing are crucial for CRNs. The paper proposes a multi-agent deep reinforcement learning (DRL) based CSS method to help SUs efficiently finding a vacant channel by the cooperation with their partners. ① Two partner selection algorithms are proposed named as Reliable Partner CSS and Adaptive Partner CSS, respectively. For the former, the partner selection is facilitated based on the historical sensing accuracy of SUs. While the latter takes the comprehensive consideration of both the reliability and geographical distribution of SUs to further improve the sensing accuracy. ② Multi-agent deep deterministic policy gradient (MADDPG) is adopted to resist the dynamically varying channels condition as well as the high-dimension solution space. With the feature of ‘centralized training and decentralized execution’, each SU learns to interact with the environment and select a vacant channel for transmission by its partial observation, which greatly reduces the communication overhead caused by the cooperative spectrum sensing. ③ Numerical simulation demonstrates the convergence and availability of the proposed algorithms. No matter Reliable Partner CSS or Adaptive Partner CSS, the sensing accuracy can be greatly enhanced comparing with other non-cooperative or centralized learning approaches. Besides, the attention mechanism is introduced to MADDPG for Adaptive Partner CSS to reveal the behavior of SUs by the visualization of attention weight, which helps to partially interpret the ‘black box’ issue of DRL.

Index Terms—Cognitive radio networks (CRNs), cooperative spectrum sensing (CSS), deep reinforcement learning, multi-agent deep deterministic policy gradient, multi-head attention.

I. INTRODUCTION

THE ever-changing wireless devices lead to a rapid growth of mobile data traffic. The massive access demand makes the spectrum extension to be more critical for wireless communication networks. However, the utilization rate of most licensed frequency bands is generally less than 30% [1], [2]. Recently, the concept of cognitive radio networks (CRNs) is proposed and envisaged as an effective technology to enhance the spectrum utilization, where the secondary users (SUs) can opportunistically reuse the spectrum already licensed to the primary users (PUs) for transmission by exploring spectrum holes spatially and temporally.

In general, there are two methods for such spectrum sharing. ① One is the *overlaying* or *interweaving* scheme [3], also known as listen-before-talk (LBT), i.e., SUs are required to sense whether one channel is available before the transmission to avoid interfering with PUs. ② The other is the *underlaying* scheme, where SUs are allowed to adjust their transmission power to guarantee that the accumulated interference experienced at PUs is less than a tolerable threshold [4].

Although the underlaying scheme has a better spectrum efficiency, it is a centralized approach because the channel state information (CSI) between transmitter of SUs and receiver of PUs should be known a prior for the power control, which inevitably increases the synchronization and communication overhead. Despite that there are false alarm and miss detection sometimes caused by channels noise and fading, the overlaying scheme is still an effective way in many existing researches [5], [6] for the distributive framework, i.e., each SU can be enforced to sense spectrum holes independently.

Take the overlaying scheme into consideration, where SUs have to explore and identify spectrum holes of the primary networks to avoid the collision among SUs and PUs. Since the sensing accuracy is susceptible to the channels fading and shadowing effect, SUs are supposed to keep sensing the licensed channels until they find an available channel or fail, which results in a low spectrum efficiency and time-wasting. Therefore, the cooperative spectrum sensing (CSS) technology is proposed to enhance the sensing quality and accuracy [7]. Given the available

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spectrum of CRNs may be wide and non-consecutive, it can be more efficient if each SU only senses a part of the spectrum once a time and shares the results with a fusion center or decision making authority to produce a more reliable judgment on the current spectrum occupation [7], [8]. On this basis, SUs can find vacant channels cooperatively for temporary transmission by their partial observation [9], [10], [11].

Although CSS has greatly enhanced the sensing accuracy of SUs, there are still some challenges. 1) The topology of CRNs changes dynamically because the wireless devices are mobile and join or leave the network from time to time [7]. 2) The occupation of frequency bands has an influence on the access strategy of SUs. 3) Last but not least, the channels occupation of PUs is hard to be predicted. Although the hidden Markov model has been used to predict the channels state in future, it highly depends on the spectrum usage distribution [12].

A. Prior Work and Challenges

By avoiding frequent interaction with pre-established infrastructures or time-wasting ‘sweep’ across the whole frequency bands, most existing literature adopt the cooperative sensing to maximize the overall throughput of system [3], [4], [5], [8], [9], [13], [14] or the sensing accuracy of SUs [6], [7], [9], [10], [12].

- Works in [7], [9], [10], [12], [13] adopt deep reinforcement learning (DRL) approaches to optimize the channels sensing in CRNs, where the channels occupation of PUs is taken as a Markov decision process (MDP) [7], [10]. The whole system is treated as a single global agent and fed with a positive reward when SUs correctly sense a vacant channel, and vice versa. By the training of neural networks, all SUs learn the occupation pattern of PUs and avoid the time-wasting bands scanning.
- Aforementioned DRL approaches can be further improved to the multi-agent domain [5], [16], [17], where each SU learns by itself as an individual agent. Multi-agent DRL enables the learning process to be implemented distributively, which facilitates the deployment of CRNs with overlaying scheme without the pre-established infrastructures for central management or coordination.
- DRL based CSS method has been proposed to make a single SU be global aware of the channels state and enhance the sensing accuracy [6], [10], [14], [15]. In specific, each SU learns the policy to find a dedicated channel with the largest probability to be vacant by fusing the sensing results with others, known as its *partners* or *neighbors*. Partner selection can be based on the detection ability [6] or historical sensing accuracy of SUs. However, the channels state fusion is implicitly operated in centralized DRL [13], [14], where the whole system is modeled as a single global agent and the channels state sensed by all SUs is taken as the input of neural networks.

Actually, most existing works resort to DRL methods for two significant advantages. 1) DRL is endowed with the non-linear approximation capability to solve complex optimization problems by training neural networks. Especially in CRNs, the occupation pattern of PUs is hard to be predicted and multiple SUs compete for limited frequency resources dynamically. 2) Compared with convex-based optimization [3], [4],

DRL can be easily extended to the distributive form, which is compatible with the expectation of CRNs which has no complex pre-established infrastructures to manage and coordinate SUs’ activities. However, there are still some issues to be solved:

1) *How to Design an Effective Algorithm for Partner Selection*: CSS enhances the channels sensing accuracy by the information sharing among SUs, which brings tremendous communication overhead. The basic idea is that each SU only shares and fuses CSI with its partners. For example, when two SUs are geographically far away, their sensed results may be meaningless to each other due to the channels fading and shadowing. Thus, an effective algorithm for partner selection is the key to improve the sensing accuracy of SUs.

2) *Value-Based Q-Learning is Unsuitable for Complex Environments*: Q-learning works poorly in complex environments [18] due to the following two reasons: 1) Each agent as a part of the environment changes all the time, which makes the environment non-stationary from the perspective of individual (Due to the lack of global observation, the channels state is time-varying and unreliable.). In consequence, agents could not be trained to cover all ergodic state and be prevented to use the past experience to replay in a straight forward way. 2) The coherent nature of error propagation makes the state and action in Q-learning to be discrete, which is infeasible with the channels belief of SUs in consecutive probability [10].

3) *DRL Lacks Interpretability*: Compared with other optimization approaches which can be proven by mathematics, such as convex optimization [3], [4], machine learning is mostly like a ‘black box’ and hardly to be understood, i.e., lack of the interpretability [19], [20]. For example, it is difficult to explain why a SU driven by machine learning makes one channels choice, why a SU depends more on one partner but less on the others, and what would happen if there is the channels condition SUs never experienced. Such doubts block DRL approaches to be completely trusted into practice, especially when multiple agents act as parts of the environment and interact with each other dynamically in CRNs.

B. Main Contributions

The differences between our work and the existing literatures are summarized in Table I, the main contributions of the paper are as follows:

- Two partner selection algorithms for CSS are proposed to enhance the sensing accuracy of SUs, i.e., *Reliable Partner CSS* and *Adaptive Partner CSS*. In specific, the channels belief fusion in Reliable Partner CSS is facilitated based on the historical sensing accuracy of SUs. Since each SU can be globally aware of the recent primary channels, the one-time success probability achieves above 85%. Since Adaptive Partner CSS is operated with the comprehensive consideration of both the reliability and geographical distribution of SUs, the sensing accuracy can be further improved.
- To tackle the mixed integer nonlinear programming (MINLP) problem with computation complexity, the policy based multi-agent deep deterministic policy gradient

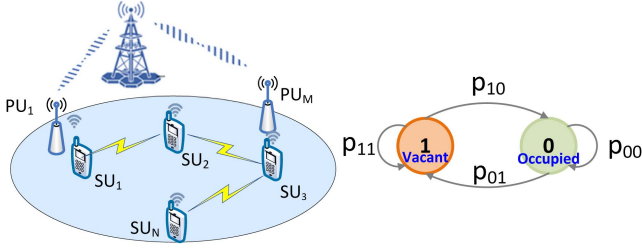


Fig. 1. System model and Markov process of channels state.

(MADDPG) network is combined with the two CSS algorithms aforementioned. Equipped with the ‘centralized training and decentralized execution’ feature, each SU can individually select a vacant channel for transmission by its partial observation, which greatly reduces the synchronization and communication overhead caused by the cooperative spectrum sensing.

- The attention mechanism is introduced to MADDPG for Adaptive Partner CSS where SUs can autonomously learn to pay more attention on the partners according to their own interests. By the visualization of attention weight, the agents’ behavior and inter-connection can be further justified and explained in a dynamic environment, which is necessary to gain the scientific insight on the working of a ‘black box’.

The rest of paper is organized as follows. Section II details the system model and formulates the optimization problem related to maximize the sensing accuracy, which can be solved by two approaches, i.e., Reliable Partner CSS and Adaptive Partner CSS detailed in Sections III and IV, respectively. The computation complexity of the proposed DRL based CSS algorithms is analyzed in Section V. The numerical results are given and discussed in Section VI and the conclusion is in Section VII.

II. SYSTEM MODEL

A. Channels Occupation of PUs

As shown in Fig. 1, there are $\mathcal{M} = \{1, \dots, M\}$ PUs coexisting with $\mathcal{N} = \{1, \dots, N\}$ SUs with $\mathcal{N}_f = \{1, \dots, N_{fb}\}$ orthogonal channels in a CRN, where SUs can opportunistically access vacant channels without interfering PUs. The transmitting power of PUs is set at a fixed value, and the channel noise is assumed to be Additive White Gaussian Noise (AWGN) with the power of δ_0^2 .

In general, PUs have the first priority to occupy all the wireless channels. Channels state can be modeled by a two-state dynamic Markov process [5], i.e., vacant and occupied. The transition probability $\mathbf{P}_f$ is given by:

$$\mathbf{P}_f = \begin{bmatrix} p_{0,0} & p_{0,1} \\ p_{1,0} & p_{1,1} \end{bmatrix}, \quad (1)$$

where $p_{1,0}$ is the transition probability of channels from a vacant state to be occupied, and vice versa. SUs can only use the channels in vacant state.

Let $\mathbf{O}[t] = [o_1[t], \dots, o_{N_{fb}}[t]]$ be the channels state vector at the t th time slot, where $o_f \in \{0, 1\}$, $1 \leq f \leq N_{fb}$. When $o_f[t] = 1$, the f th channel is vacant, while $o_f[t] = 0$ means the

channel is currently occupied by a PU. Assume that channels follow the quasi-static block fading, i.e., channels state remains unchanged in a transmission frame. SUs need to keep sensing channels one by one until find a vacant channel. The access is considered to be failed if none vacant channels can be found after a certain number of sensing. This procedure is time-wasting especially when the number of channels or frequency bands is significantly larger than the number of SUs [10].

B. Energy Detection of SUs

Since the base station is supposed to manage the channels assignment for licensed users, the channels collision among PUs can be effectively avoided. Define P_m be the transmission power of PU_m , $h_{m,n}$ ($1 \leq n \leq N$) be the channels gain between PU_m and SU_n , and $x_{m,f} \in \{0, 1\}$ be the transmitted symbol of PU_m at the f th channel. Then the received signal in the f th channel at SU_n can be expressed as:

$$y_{n,f} = \sum_{m=1}^M x_{m,f} P_m h_{m,n} + v, \quad (2)$$

where $v \sim \mathcal{GN}(0, \delta_0^2)$.

Suppose that SUs adopt the energy detection technology to sense the occupation of PUs at a specific channel. The average received energy of SU_n at the f th channel can be written as:

$$u_{n,f} = \frac{1}{N_{\text{ED}}} \sum_{i=1}^{N_{\text{ED}}} |\bar{y}_{n,f}^i|^2, \quad (3)$$

where $|\bar{y}_{n,f}^i|^2$ is the i th sampled energy at the f th channel and N_{ED} is the number of samples to be employed for energy detection. Therefore, the sensing accuracy of SU_n , i.e., the probability of correctly sensing a vacant channel can be denoted as:

$$p_n(u_{n,f} \leq \Gamma_n | o_f = 1), \quad (4)$$

where Γ_n is the energy detection threshold of SU_n .

C. Problem Formulation

To enhance the sensing accuracy and avoid the inefficient spectrum sweep, SUs are expected to only sense the channel with the largest probability to be vacant. Let $a_n[t] \in \mathcal{N}_f^+$ ($\mathcal{N}_f^+ = \{\mathcal{N}_f \cup 0\}$) be the channel to be sensed at the t th time slot, where $a_n[t] = 0$ means that SU_n does not sense any channels and keeps silent in the following access stage till the next time slot. The *one-time success probability*, i.e., the sensing accuracy in (4), of SU_n can be rewritten as:

$$p_n(u_{n,f} < \Gamma_n | a_n = f, o_f = 1). \quad (5)$$

The optimization objective of each SU is to maximize the expected probability for correctly sensing a vacant channel:

$$\text{P1 : } \max_{a_n[t]} \min \mathbb{E}[p_n], \quad (6)$$

$$\text{s.t.: } o_f[t] \sim \text{MDP}(\mathbf{P}_f, o_f[t-1]). \quad (\text{C1})$$

$$a_n[t] \in \mathcal{N}_f^+, \forall n, \quad (\text{C2})$$

$$\sum \mathcal{I}_f(a_n[t]) \leq 1, \forall n, f. \quad (\text{C3})$$

where $\mathcal{I}_f(\cdot) \in \{0, 1\}$ is an indication function, i.e., $\mathcal{I}_f(a_n[t] = f) = 1$, otherwise $\mathcal{I}_f = 0$.

Constraint (C1) implies that the channels state subjects to a two-state dynamic Markov process. Constraint (C2) denotes all the possible options for the channel sensing of SUs. While constraint (C3) limits that a specific channel f can only be sensed by one SU at most in a time slot, which effectively avoids the collision when multiple SUs try to access the same channel simultaneously.

Since $\mathcal{I}(\cdot)$ is a binary indication function and the variable $a_n[t]$ is in discreteness, P1 is a MINLP problem, which is hard to obtain the closed-form solution in polynomial time. Besides, the channels occupation of PUs varies from time to time, the prior CSI can not be perfect, which leads to unaffordable computation complexity when a mass of users are involved in the system¹.

Thanks to the non-linear approximation ability of deep neural networks [21], the model-free feature of DRL, such as Q-learning [11], DQN [10], and DDPG [22], makes it easier to map the high dimensional state space to a specific action without the convexity requirement for objective or reward functions, i.e., SUs can learn to find a channel with the largest probability to be vacant from their historical experience.

MADDPG [18] scheme which inherits DDPG method into the multi-agent domain is adopted in the paper where each SU is taken as an independent agent to interact with the environment. It not only eliminates the non-stationary feature of Q-learning and policy gradient, but also reserves the great advantage of DDPG that be able to search the action space in a continuous rather than discrete way. More important, with the feature of ‘*centralized training and decentralized execution*’, the communication overhead for information synchronization can be greatly reduced.

D. MADDPG Based Partner Selection for CSS

As shown in Fig. 2, two MADDPG based partner selection algorithms are proposed for CSS to enhance the sensing accuracy of SUs, i.e., Reliable Partner CSS detailed in Section III and Adaptive Partner CSS detailed in Section IV.

- **Reliable Partner CSS:** Shown by the blue diagram, each SU selects its partners and makes the channels state fusion based on the historical reliability. On this basis, MADDPG is adopted to enable each SU make the sensing action independently with a shared and simple reward.
- **Adaptive Partner CSS:** Shown by the yellow diagram, SUs take the comprehensive consideration of both the reliability and geographical distribution to order the preference of partners. It is more trackable for each SU to adaptively pay differentiated attention to partners according to its own interests quantized by the introduction of attention weight.

¹Assume that the number of channels shared by each PU is $N_m \in [0, N_{fb}]$. Thus for each SU, $a_{n,f}[t]$ has $(N_m + 1)$ dimension. Given there are three possible state for each channel, i.e., PU transmission, SU transmission and vacancy [14], the system complexity should be $\mathcal{O}(3^{\sum_{m=1}^M N_m})$. Because of the existing of sensing errors, there are at most $\prod_{m=1}^M (N_m + 1)$ possible sensing combination for all SUs in each time slot. As a result, the solution space for P1 works $\mathcal{O}(3^{\sum_{m=1}^M N_m} \prod_{m=1}^M (N_m + 1))$, which is obviously high-dimensional.

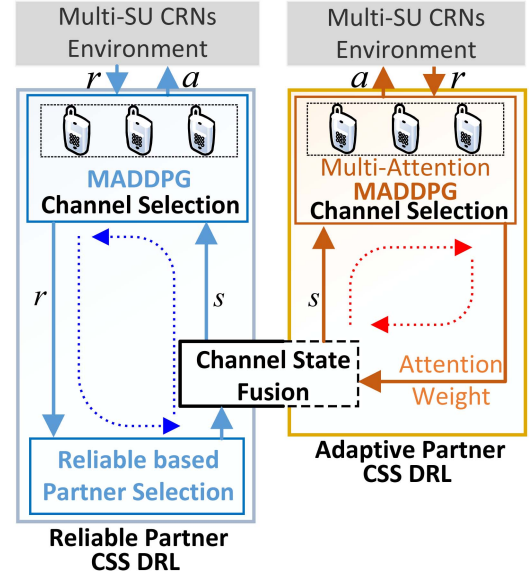


Fig. 2. MADDPG based Reliable and Adaptive Partner CSS.

III. MADDPG BASED RELIABLE PARTNER CSS

In general, there are three phases for MADDPG based Reliable Partner CSS:

- **Partner Selection:** Each SU selects its partner set $\mathcal{G}_n$ based on the historical sensing accuracy of other SUs. The one who has better one-time success probability in the past is regarded more reliable.
- **Channels Belief Fusion:** SUs dynamically adjust their partners' weight $\omega_{n,i}$, $i \in \mathcal{G}_n$ according to the reliability, which can be further weighted-sum as the channels belief by (12).
- **Action Mapping:** The channels belief $b_{n,f}$ is mapped to a proper sensing action a_n by MADDPG.

A. Reliable Partner Selection

Due to the variant location distribution of SUs and the time-varying of channels, the sensing ability of SUs changes dynamically and is unknown to others. Thus it is reasonable to set partners different weight $\omega_{n,i}$ according to their reliability.

In the t th time slot, when SU_i acts as the partner of SU_n and leads to a larger historical accumulative reliability, it is assigned to a higher weight $\omega_{n,i}$. Each SU sorts and updates its partner set $\mathcal{G}_n$ based on such reliability. The partner selection algorithm related to the sensing reliability and new partners exploring [6] is designed as follows:

- If SU_i helps SU_n to attain a better reliability in the past t^s time slots, it shall be selected as the partner of SU_n in priority. Let $\rho_{n,i}$ be the reliability of SU_i to SU_n :

$$\rho_{n,i}[t] = \sum_{t'=1}^{t^s} \gamma^{t-t'} r_n[t-t'] \mathcal{I}\{i[t-t'] \in \mathcal{G}_n \cup n\}, \quad (7)$$

where $\mathcal{G}_n$ is the cooperative partner set, the subscript $\cup n$ denotes the current SU itself, and $\gamma < 1$ is the discount factor. If SU_n selects SU_i as its partner at the t th time slot, there is $\mathcal{I}\{\cdot\} = 1$, $i[t] \in \mathcal{G}_n$. Otherwise $\mathcal{I}\{\cdot\} = 0$.

- SU who is outside of the partner set $\mathcal{G}_n$ still has the probability to be selected as a new partner. Let $\eta_{n,i}$ be the number of times that SU_{*i*} becomes the partner of SU_{*n*} in the past t^s time slots:

$$\eta_{n,i}[t] = \sum_{t'=1}^{t^s} \mathcal{I}\{i[t-t'] \in \mathcal{G}_n\}. \quad (8)$$

SU with a larger historical reliability or less cooperative times shall be more probable to be selected as a new partner, which means that the partner selection is proportional to the reliability $\rho_{n,i}$ but inverse proportional to $\eta_{n,i}$. Thus, $\mathcal{G}_n$ can be updated by:

$$\mathcal{G}_n[t] = \arg_{i \in \{\mathcal{N}\}} \text{sortmax}_{N_{\text{coop}}} \left(\rho_{n,i} + \epsilon \frac{1}{\eta_{n,i}} \right), \quad (9)$$

where the function $\text{sortmax}_{N_{\text{coop}}}(\cdot)$ returns N_{coop} number of index as descent order of $(\cdot)$, $0 \leq N_{\text{coop}} \leq N - 1$ and $\epsilon < 1$ is the exploring factor.

- Based on the reliability of SUs in the set of $\{\mathcal{G}_n \cup n\}$, the fusion weight of SU_{*i*} to SU_{*n*} can be calculated by:

$$\omega_{n,i}[t] = \frac{\rho_{n,i}[t]}{\sum_{i' \in \{\mathcal{G}_n \cup n\}} \rho_{n,i'}[t]}, \omega_{n,i} \in [0, 1]. \quad (10)$$

B. Belief Fusion and Decay

Once SUs have sensed a channel no matter occupied or vacant, they maintain such results as a set of channels state. In other words, the state of a specific channel f is deterministic once it has just been sensed and is saved in the vector $\bar{\mathbf{b}}_n(t) = [\bar{b}_{n,1}, \dots, \bar{b}_{n,N_{fb}}]$, where $\bar{b}_{n,f} \in \{0, 1\}$ denotes the original channels state sensed by SU_{*n*}, i.e., $\bar{b}_{n,f} = 1$ for vacancy and $\bar{b}_{n,f} = 0$ for occupation.

$$\bar{b}_{n,f}(f = a_n[t]) = \begin{cases} 1, & a_n[t] \text{ channel is vacant,} \\ 0, & a_n[t] \text{ channel is occupied.} \end{cases} \quad (11)$$

However, such state becomes uncertain along with time because the channels occupation of PUs changes dynamically. Correspondingly, the reliability gradually decays with time until 50% certainty to be vacant or occupied (totally unknown), which means that $\bar{b}_{n,f}$ decays to 0.5 if the channel is not sensed in the adjacent time slot [10]. So the decay rate can be expressed by a forgetting factor $\alpha \in [0.5, 1]$. If $\bar{b}_{n,f} > 0.5$, SUs tend to believe that channel f is vacant and vice versa. If $\bar{b}_{n,f} = 0.5$, the channels state is totally uncertain.

As aforementioned, SUs are endowed with the global awareness of channels state by cooperating with each other. As a result, when SU is uncertain about a channel or even misses to sense, it can still perform the channel estimation through the shared information from other SUs. To eliminate the global broadcasting which may lead to huge communication overhead, assume that one SU only receives such shared information from its partners. Since multiple partners may share the information about the same channel, the state of channel f at SU_{*n*} should be fused as a *weighted-sum*:

$$b_{n,f} = \sum_{i \in \{\mathcal{G}_n \cup n\}} \omega_{n,i} \bar{b}_{i,f}, \quad (12)$$

Algorithm 1: Channels Belief Decays With Time and Fuses With Partners.

```

1 for  $f = 1$  to  $N_{fb}$  do
2   if  $a_n(t) = f$  then
3     if  $o_f(t) = 1$  (band  $f$  is found vacant) then
4        $\bar{b}_{n,f} = 1$ ;
5     else
6        $\bar{b}_{n,f} = 0$ ;
7     end
8   else
9     if  $\bar{b}_{n,f} \geq 0.5$  then
10       $\bar{b}_{n,f} = \max\{0.5, \alpha b_{n,f}\}$ 
11    else
12       $\bar{b}_{n,f} = \min\{0.5, 1 - \alpha(1 - b_{n,f})\}$ 
13    end
14  end
15  For Reliable Partner CSS, channels belief  $\mathbf{b}_n(t)$  is
    calculated according to Eq. (12), which detailed
    in Sec. III-B.
16  For Adaptive Partner CSS, channels belief  $\mathbf{b}_n(t)$  is
    calculated according to Eq. (29), which detailed
    in Sec. IV-C.
17   $f \leftarrow f + 1$ 
18 end

```

where $b_{n,f} \in [0, 1]$ is the fused channels belief of SU_{*n*} (denoted as *channels belief* for briefness), and $\omega_{n,i} \in [0, 1]$ is the reliability dependent weight. For each SU, the channels belief vector $\mathbf{b}_n(t) = [b_{n,1}, \dots, b_{n,N_{fb}}]$ fades with time and can be fused with its partners according to Algorithm 1.

C. Multi-Agent DDPG Based Deep Learning

By adopting the actor-critic (AC) framework, each SU in MADDPG acts as an independent intelligent agent to interact with the environment, i.e., try to find the channel with the maximum probability to be vacant from their historical experience. Once well trained, SUs can take the vacant channel for immediate secondary communication without scanning the whole bands or retrying many times.

- *State*: Each SU maintains a set of channels belief as the state, i.e. $\mathbf{s}_n[t] = [b_{n,1}, \dots, b_{n,N_{fb}}]$, where $b_{n,f}$ is calculated according to (12).
- *Action*: When SU_{*n*} initiates a call, it makes a sensing action $a_n[t] \in \mathcal{N}_f^+$ to seek a vacant channel, i.e., determines which channel to sense at the t th time slot.
- *Reward*: Reward of a single agent $r_n[t]$ is returned from the environment to criticize the action $a_n[t]$ according to (13). In specific, an agent who successfully senses a vacant channel gets a positive reward '1', and otherwise a negative value '-1'.

$$r_n(\mathbf{s}_n, a_n) = \begin{cases} 1, & a_n[t] \text{ channel is vacant,} \\ -1, & a_n[t] \text{ channel is occupied.} \end{cases} \quad (13)$$

Since each SU in MADDPG works as an independent agent and pursuits to maximize its own reward, the optimization

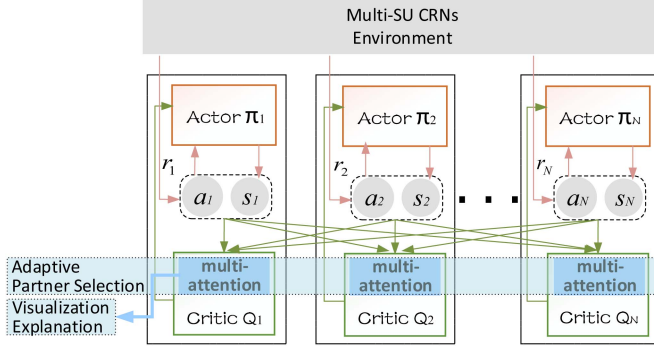


Fig. 3. Actor-Critic framework of MADDPG for deep learning.

problem P1 can be further written as:

$$\begin{aligned} \text{P2: } \max_{a_n[t]} \mathbb{E} \left[\sum_{t'=0}^{\infty} \gamma^{t'} r_n[t+t'] \right], \\ \text{s.t.: (C1) - (C3).} \end{aligned} \quad (14)$$

As shown in Fig. 3, each agent has an actor network and a critic network. Actor π_n maps the local state s_n into a proper action a_n according to its network parameters μ_n , which is also known as the policy of actor π_n . During the training, actor π_n generates a deterministic policy instead of the policy gradient, i.e., choosing a random action from a determined distribution. Critic Q_n takes the global action $\mathcal{A}[t] = [a_1[t], \dots, a_N[t]]$ and global state $\mathcal{S}[t] = [s_1[t], \dots, s_N[t]]$ as the inputs, and evaluates the performance of policy μ_n by outputting the policy gradient of actor π_n .

It should be noticed that although the critic Q_n takes the global action-state pairs $(\mathcal{S}[t], \mathcal{A}[t])$ to generate a more robust evaluation for the actor, it only works during the training stage. During the execution stage, the well-trained actor π_n makes the optimal action $a_n[t]$ independently based on its local channels belief $s_n[t]$ without interfering with other agents. Such feature is known as ‘*centralized training and decentralized execution*’, which greatly reduces the communication overhead for information synchronization. In specific, in the 17th line of Algorithm 2, when “*training = True*”, each agent takes the global action $\mathcal{A}$, global state $\mathcal{S}$ and local reward r_n to update the corresponding actor and critic network. However, only the local state s_n is needed to generate the action a_n during the execution.

Besides, both the actor and critic have double sub-networks, i.e., an online network and a target network, which is a common technology to ensure the learning stability and overcome the overoptimism in large-scale problems [22]. MADDPG further adds noise $\mathbb{N}$ to explore better strategies in the training stage. Another essential technique is the experience replay buffer (RB). Each agent is equipped with a RB to store the (state, action, reward, next state) transition $\{s[t], a[t], r_n[t], s[t+1]\}$, which can be fetched out randomly to update the parameters. Together with mini-batch, experience replay can effectively avoid the highly correlation of successive updating.

By experience replay and deep neural network (DNN) for training, the policy-based DRL updates parameters μ_n in the

gradient direction to enlarge the value function J :

$$J(\mu_n) = \sum_{s_n \in \mathbb{S}} d^\pi(s_n) \sum_{a_n \in \mathbb{A}} \pi(s_n | \mu_n) Q(\mathcal{S}, \mathcal{A} | \theta_n), \quad (15)$$

where $d^\pi(s_n)$ is the stationary state distribution function based on policy μ_n and thus $\pi(s_n | \mu_n)$ produces a deterministic action a_n based on state s_n . The actor network parameters μ_n and the critic network parameters θ_n can be approximated by DNN and are mutual dependent in the learning process. For notation brevity, we take the subscript to denote the agent number and the superscript to denote the step number unless otherwise stated.

The interaction with environment and the process of training networks are listed as follows:

- The online actor network tries to map the current state s_n^t to an action $a_n^t = \pi_n(s_n^t | \mu_n)$. Parameters μ_n can be updated by:

$$\mu_n^{t+1} = \mu_n^t + \alpha_\mu \nabla_{\mu} J(\mu_n), \quad (16)$$

where α_μ is the step-size for parameters updating and $\nabla_{\mu} J(\mu_n)$ is the corresponding policy gradient of π_n .

- The target actor network $\pi'_n(s_n^{t+1} | \mu'_n)$ predicts the action of the next state s_n^{t+1} from either the environment or experience RB with parameters μ'_n :

$$a_n^{t+1} = \pi'_n(s_n^{t+1} | \mu'_n). \quad (17)$$

- The online critic network $Q_n(\mathcal{S}, \mathcal{A} | \theta_n)$ updates the policy gradient $\nabla_{\mu} J(\mu_n)$ by (18):

$$\begin{aligned} \nabla_{\mu} J(\mu_n) = \mathbb{E} [\nabla_{a_n^t} Q(\mathcal{S}, \mathcal{A} | \theta_n)_{\mathcal{S}=\mathcal{S}^t, \mathcal{A}^t=[\pi_1(s_1^t), \dots, \pi_N(s_N^t)]}] \\ \nabla_{\mu} \pi_n(s_n^t | \mu_n), \end{aligned} \quad (18)$$

where the input pairs $(\mathcal{S}, \mathcal{A})$ are selected from the mini-batch. The goal of the online critic network is to minimize the temporal difference (TD) error L_n by training parameters θ_n .

$$L_n(\theta_n) = \frac{1}{\mathbb{M}} \sum_t [(y_n^t - Q_n(\mathcal{S}^t, \mathcal{A}^t | \theta_n))^2] \quad (19)$$

$$\theta_n^{t+1} = \theta_n^t + \alpha_\theta \nabla_{\theta} Q_n(\mathcal{S}^t, \mathcal{A}^t | \theta_n), \quad (20)$$

where $\mathbb{M}$ is the mini-batch size, y_n^t is calculated by the target critic network and α_θ is the step-size for parameters updating.

$$y_n^t = r_n^t + \gamma Q'_n(\mathcal{S}^{t+1}, \mathcal{A}'^{t+1} | \theta'_n) |_{a_n^{t+1} = \pi'_n(s_n^{t+1})}, \quad (21)$$

where r_n^t is the reward of action a_n^t by interacting with the environment, a_n^{t+1} is the output of the target actor network based on (17) and $\mathcal{A}'^{t+1}$ is the collection of other agents' action sampled from RB combining with a_n^{t+1} , i.e., $\mathcal{A}'^{t+1} = [a_1^{t+1}, \dots, a_n^{t+1}, \dots, a_N^{t+1}]$.

- The target critic network $Q'_n(\mathcal{S}^{t+1}, \mathcal{A}'^{t+1} | \theta'_n)$ updates the target critic value y_n^t with the inputs $(\mathcal{S}^{t+1}, \mathcal{A}'^{t+1})$.
- For both the target actor and critic network, the parameters are soft-updated as follows:

$$\begin{cases} \mu'_n = \leftarrow \iota \mu_n + (1 - \iota) \mu'_n, \\ \theta'_n = \leftarrow \iota \theta_n + (1 - \iota) \theta'_n, \end{cases} \quad (22)$$

where ι is the forgetting factor.

IV. ADAPTIVE PARTNER CSS WITH ATTENTION CRITIC

Because the channels fading and shadowing effect may increase the signal uncertainty at receivers, the sensing result in Reliable Partner CSS may lose the reference value when SUs are geographical far away. For example, when a partner of SU senses a channel occupied, SU may receive an error information for vacancy. Therefore, SUs should pay more attention to the partners nearby for less fading and shadowing.

Moreover, one critical limitation of DRL is the lack of interpretability from human's perspective. Recently, multi-head attention has achieved great success in sequence transduction tasks [23], e.g. the inter-connection among agents [24] or the context comprehension in natural language processing (NLP) [25] can be well interpreted by the analysis of attention weight. Inspired by this, the attention mechanism is introduced to MADDPG to reveal the deep connection of SUs during the action mapping and explain their mutual impact in a broader range.

Hence, Adaptive Partner CSS with attention critic is further proposed in this section. Shown as the blue blocks in Fig. 3, a multi-head attention layer is added to the critic network of MADDPG named as *Attention Critic*, which aims at modeling the partners' policy and adjusting its own policy [26]. The attention weight $W_n(\mathcal{S}, \mathcal{A})$ can be interpreted as the impact of other agents to SU_n and further used for the channels state fusion. Therefore, although the partner selection may not such explicit in the learning process, the relationship among SUs can be revealed by the visualization of attention weight.

A. Deep Learning for Adaptive Partner CSS

The definition of state, action and reward for adaptive partner selection can be detailed as follows:

- *State*: The channels belief is taken as the state of SU_n , i.e., $s_n[t] = [b_{n,1}, \dots, b_{n,N_{fb}}]$, where $b_{n,f}$ is now calculated according to (29) discussed later.
- *Action*: SU_n makes the sensing action $a_n[t] \in \mathcal{N}_f^+$ to seek a vacant channel.
- *Reward*: Equipped with the awareness of the geographical distribution of SUs, Adaptive Partner CSS takes the negative value of detected energy $u_{n,f}[t]$ as the reward rather than an integer '1' or '-1' in (13).

$$r_n(s_n, a_n) = -u_{n,a_n}. \quad (23)$$

The detected energy can be instantaneously obtained at SUs' receiver rather than determining the channels occupation after a successful transmission. The smaller detected energy sensed at the selected channel, the larger probability to be vacant.

Shown as Fig. 3, each agent in Adaptive Partner CSS holds the same actor network with that in Reliable Partner CSS. Although the agents in both Reliable Partner CSS and Adaptive Partner CSS have the same inputs for the critic network, i.e., the global state $\mathcal{S}$ and global action $\mathcal{A}$, each agent in the latter would like to pay more attention to SUs following its own interests, which can be automatically learned by the attention critic.

Algorithm 2: MADDPG Based CSS Algorithm.

```

1 for  $t = 1, \dots, \text{Max episode}$  do
2    $SU_n$  outputs a sense action  $a_n^t$  as the input of  $s_n^t$ 
   according to its actor network  $\pi_n(s_n|\mu_n)$ ;
3    $SU_n$  interacts with the environment, i.e., tries to
   access  $a_n^t$  channel, and updates its local state of
   channel  $b_{n,a_n^t}$  according to Algorithm 1;
4   if Reliable Partner CSS then
5      $SU_n$ 's reward  $r_n^t$  is obtained by Eq. (13);
6     for  $i = 0$  to  $N, i \neq n$  do
7       Calculate the reliability  $\rho_{n,i}^t$  for  $SU_i$  as
       Eq. (7);
8       Update  $\mathcal{G}_n$  as Eq. (9);
9       Update reliability weight  $\omega_{n,i}$  according to
       Eq. (10);
10    end
11    According to Eq. (12), fuse the channels state
    with partners in  $\mathcal{G}_n$  to construct  $s_n^t$ ;
12  end
13  if Adaptive Partner CSS then
14     $SU_n$ 's reward  $r_n^t$  is obtained by Eq. (23);
15    According to Eq. (29), fuse the channels belief
    with attention weight  $\tilde{\omega}_{n,i}$  to construct  $s_n^t$ ;
16  end
17  if training = True then
18    Collect the global action  $\mathcal{A}^t = [a_i^t]$ , global state
     $\mathcal{S}^t = [s_i^t]$ , where  $i \in \mathcal{N}$ ;
19    Store transition  $[\mathcal{S}^t, \mathcal{A}^t, r_n^t, \mathcal{S}^{t+1}]$  in RB ;
20    Sample a random mini-batch of  $\mathbb{M}$  from RB;
21    Calculate  $y_n^t$  by Eq. (21);
22    Update critic online by minimizing Eq. (19);
23    Update the actor online policy by Eq. (16);
24    Update the actor and critic target by Eq. (22);
25    if Adaptive Partner CSS then
26      Update the attention weight  $\tilde{\omega}_{n,i}$  according
      to Eq. (27)
27    end
28  end
29   $t \leftarrow t + 1$ 
30 end

```

The critic of SU_n in Adaptive Partner CSS can be written as $Q_n^c(\mathcal{S}, \mathcal{A})$, which aims for the estimation of $\mathbb{E}_{\mathcal{S}, \mathcal{A} \sim \rho(\cdot)}[Q_n^c(\mathcal{S}, \mathcal{A})]$ over the behavior distribution² $\rho(\cdot)$.

$$\begin{aligned}
Q_n^c(\mathcal{S}, \mathcal{A}) &= \mathbb{E}_{\mathcal{S}, \mathcal{A} \sim \rho(\cdot)}[Q_n^c(\mathcal{S}, \mathcal{A})] \\
&= \sum_{i, a_i \in \mathcal{A}} p_i(a_i|\mathcal{S}) Q_n^i(s_i, a_i|\mathcal{S}, \mathcal{A}), \quad (24)
\end{aligned}$$

where $p_i(a_i|\mathcal{S})$ is the probability of $a_i \in \mathcal{A}$ in the condition of global observation $(\mathcal{S}, \mathcal{A})$, and $Q_n^i(i \neq n)$ is the conditional Q-value of other agent. For notation brevity, the subscript of Q_n^i denotes the current agent SU_n and the superscript denotes other agent $i \in \mathcal{N}$.

² $\rho(\mathcal{S}, \mathcal{A})$ is a probability distribution over global state $\mathcal{S}$ and global action $\mathcal{A}$ referred as the behavior distribution [27].

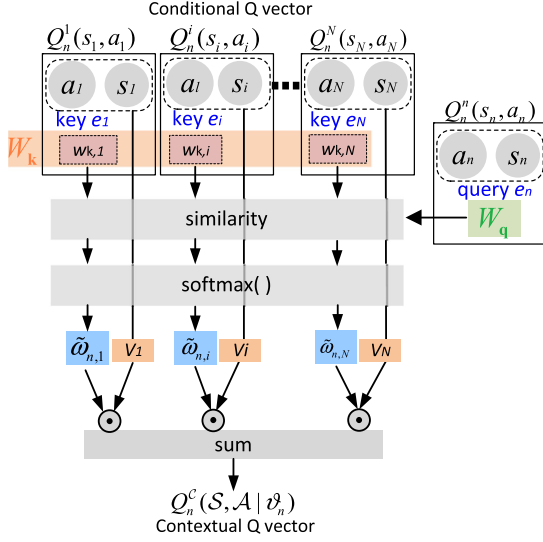


Fig. 4. Framework of one attention head.

The probability $p_i(a_i|\mathcal{S})$ is required for calculating the expectation of Q_n^c . However, it is hard to approximate. For a dedicated global state $\mathcal{S}$, agent SU_i may take action a_i with different probabilities depending on its own policy. Meanwhile, such policy changes from time-to-time when multiple agents learn concurrently.

The attention mechanism is introduced to approximate all $p_i(a_i|\mathcal{S})$ jointly at agent SU_n by a weight vector $\tilde{\mathbf{W}}_n(\vartheta_n) = [\tilde{w}_{n,1}(\vartheta_n), \dots, \tilde{w}_{n,N}(\vartheta_n)]$. Thus, the attention critic of agent SU_n can be rewritten as $Q_n^c(\mathcal{S}, \mathcal{A}|\vartheta_n)$ with the network parameters ϑ_n , which can be treated as the weighted summation of other agents' Q-value and thus named as 'contextual Q-value' to guide the parameters updating for the actor.

B. Attention Critic for MADDPG

With the help of attention critic network, each agent can tend to more important partners measured by the attention weight $\tilde{\mathbf{W}}_n$, which implies that all the other SU_i can automatically be the partners of the SU_n but with different attention weights. As shown in Fig. 4, Each partner's action and state contained in Q_n^i can be encoded into the contextual Q-value Q_n^c adaptively according to the normalized weight.

- $Q_n^i(s_i, a_i|\mathcal{S}, \mathcal{A})$ of SU_i is represented by $G_i(s_i, a_i)$, where $G_i(\cdot)$ is a basic fully-connected multi-layer perceptron (MLP) network that encodes (s_i, a_i) into $G_i(s_i, a_i)$ [28]. Then, $G_i(s_i, a_i)$ is further transformed into e_i named as 'key', while $Q_n^n(s_n, a_n|\mathcal{S}, \mathcal{A})$ of the current agent SU_n can be transformed into e_n named as 'query'.
- Attention weight $\tilde{w}_{n,i}$ is generated by comparing the similarity between the key e_i and a given query e_n through a similarity function $f(\cdot)$. In general, the softmax function is used to get the normalized attention weight. Thus attention weight can also be considered as a probability distribution due to $\sum_{i=1}^N \tilde{w}_{n,i} \equiv 1$.

$$\tilde{w}_{n,i} = \text{softmax}(f(e_i, e_n)) = \frac{\exp(f(e_i, e_n))}{\sum \exp(f(e_i, e_n))}. \quad (25)$$

- Contextual Q-value can be calculated as a weighted summation of $Q_n^i(s_i, a_i|\mathcal{S}, \mathcal{A})$. In general, each attention head probably only focuses on one aspect among agents [29], which means SU_n could adaptively attend to conditional Q_n^i by its own interests.

$$Q_n^c(\mathcal{S}, \mathcal{A}|\vartheta_n) = \sum_{i=1}^N \tilde{w}_{n,i} Q_n^i(s_i, a_i|\mathcal{S}, \mathcal{A}). \quad (26)$$

C. Key Implementation and Belief Fusion

Since $G_i(\cdot)$ is transformed into keys and queries by the parameter matrix W_k and W_q with the dimension of L , (25) can be rewritten as:

$$\tilde{w}_{n,i} = \text{softmax}(e_n^T W_q^T W_k e_i). \quad (27)$$

Value v_i of agent SU_i is obtained by passing $G_i(s_i, a_i)$ through a transform matrix V , and the contextual Q-value Q_n^c of agent SU_n can be written as a weighted summation of all the agent's value. Thus (26) can be transferred to:

$$Q_n^c(\mathcal{S}, \mathcal{A}|\vartheta_n) = H\left(\sum_i \tilde{w}_{n,i} v_i\right) = \sum_{i=1}^N \tilde{w}_{n,i} H(V G_i(s_i, a_i)), \quad (28)$$

where $H(\cdot)$ is also a MLP network to decode $\sum_i \tilde{w}_{n,i} v_i$ back into a $(N_f + 1)N$ dimensional vector in accordance with $Q_n^c(\mathcal{S}, \mathcal{A})$.

Each attention head uses the parameter set (W_k, W_q, V) and similarity function $f(\cdot)$ to give rise to the contribution from agents SU_i to the current agent SU_n , i.e., the attention weight $\tilde{w}_{n,i}$ learned from critic. Thus the channels belief can be fused by the attention weight:

$$b_{n,f} = \sum_{i=1}^N \tilde{w}_{n,i} \bar{b}_{i,f}. \quad (29)$$

Notice that although the attention layer is embedded in MADDPG, it only differs in the generation of Q-value. The inputs and outputs of the critic network are identical with those discussed in Section III-C. So the parameters updating still follows (18), (20) and (22).

V. COMPLEXITY ANALYSIS

The computation complexity of DRL is in terms of the floating point of operations (FLOPs) [30]. No matter Reliable Partner CSS or Adaptive Partner CSS, each SU agent consists of two isomorphic neural networks, i.e., the target network and the online network constructed by basic fully-connected MLP.³ For notation brief, MADDPG for Reliable Partner CSS and Adaptive Partner CSS are denoted by MADDPG-MLP and MADDPG-ATT respectively, unless otherwise stated.

Let u_i be the number of neurons in the i^{th} layer. For a fully-connected layer of MLP with u_i neurons as the input and u_{i+1}

³The paper mainly focuses on improving the sensing accuracy of SUs and the interpretability of machine learning. Thus although other types of networks, such as CNN, LSTM, et al., may also available, for easy-to-access, we would like to limit our contributions in the current networks configuration without trying more types of neural networks.

neurons as the output, the dot product of FLOPs computation from the i th to the $(i+1)$ th layer is $(2u_i - 1) \times u_{i+1}$, i.e., multiply u_i times and add $(u_{i+1} - 1)$ times for each neuron.

Let κ be the corresponding parameter determined by the type of activation function. For example, the Sigmoid function has $\kappa_{\text{Sigmoid}} = 4$ FLOPs because the function $\delta(z) = 1/(1 + e^{-z})$ has four mathematical operations, i.e., division, summation, exponentiation and subtraction, and each of them needs one FLOPs. Similarly, the ReLU function has $\kappa_{\text{Relu}} = 1$ FLOPs.

Thus, the computation complexity of MADDPG-MLP is:

$$\begin{aligned} & 2 \sum_{i=0}^{I-1} ((2u_{a,i} - 1)u_{a,i+1} + \kappa_i u_{a,i}) \\ & + 2 \sum_{j=0}^{J-1} ((2u_{c,j} - 1)u_{c,j+1} + \kappa_j u_{c,j}) \\ & = \mathcal{O} \left(\underbrace{\sum_{i=0}^{I-1} u_{a,i} u_{a,i+1}}_{\text{Computation FLOPs of Actor}} + \underbrace{\sum_{j=0}^{J-1} u_{c,j} u_{c,j+1}}_{\text{Computation FLOPs of Critic}} \right), \end{aligned} \quad (30)$$

where I, J are the number of layers in the actor and critic network, $u_{a,i}, u_{c,j}$ are their corresponding number of neurons in each layer respectively, and $i \in \{0, \dots, I\}, j \in \{0, \dots, J\}$. In specific, the complexity of the actor and critic network is $\mathcal{O}(N)$ and $\mathcal{O}((N_{fb} + 1)N)$, respectively.

The computation complexity of attention critic is dominated by $G(\cdot)$ and $H(\cdot)$ [31] which are constructed by MLP in the paper. The neurons number of attention layer is:

$$\begin{aligned} & \left[\underbrace{\text{encoder input}}_{(N_{fb} + 1) \times N} + \underbrace{\text{encoder output}}_{L \times N} \right] \\ & + \left[\underbrace{\text{decoder input}}_{(L \times N)N_{\text{head}}} + \underbrace{\text{decoder output}}_{(N_{fb} + 1) \times N} \right], \end{aligned} \quad (31)$$

where the first term is the neurons number of encoder $G(\cdot)$, the second term is that of decoder $H(\cdot)$, and N_{head} is number of attention heads.

Thus the complexity of attention critic is the continued multiplication from input to output, i.e., the encoder and decoder has the complexity of $\mathcal{O}((N_{fb} + 1)LN^2)$ and $\mathcal{O}(N_{fb} + 1)N^2LN_{\text{head}}$, respectively. With the consideration that L and N_{head} are the independent hyper-parameters with respect to channels number N_{fb} and SUs number N , the critic in MADDPG-ATT has the complexon of $\mathcal{O}(N_{fb}N^2)$.

In summary, during the training stage, the complexity of MADDPG-MLP and MADDPG-ATT is $\mathcal{O}(N) + \mathcal{O}(N_{fb}N)$ and $\mathcal{O}(N) + \mathcal{O}(N_{fb}N^2)$, respectively. However, such complexity are both reduced to $\mathcal{O}(N)$ for the distributed execution.

VI. NUMERICAL SIMULATIONS AND ANALYSIS

Consider the scenario that there are 6 SUs and 10 PUs randomly distributed in a rectangular area of $200 \text{ m} \times 200 \text{ m}$, and there are $N_{fb} = 10$ orthogonal channels within a bandwidth of

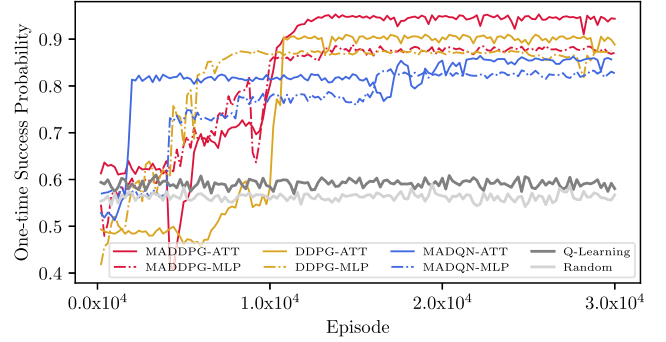


Fig. 5. One-time success probability with different algorithms.

1 MHz. The transmission power of PUs is set to be 40 mW, the average noise power density is -174 dBm/Hz and the path loss exponent is 2. The discount factor and exploring factor are $\gamma = 0.9$ and $\epsilon = 0.8$ respectively. The transition probability of each channel is uniformly distributed as $p_{1,1} \in [0.7, 1]$ and $p_{0,0} \in [0, 0.3]$, where $p_{1,0} = 1 - p_{1,1}$ and $p_{0,1} = 1 - p_{0,0}$. The parameters selection depends on the fact that the utilization of licensed spectrum is low [5]. The actor and critic neural network configuration of MADDPG are listed in Table II and RB is 10 KB.

A. Convergence and Effectiveness

Fig. 5 depicts the convergence of different DRL based CSS algorithms in terms of the one-time success probability, i.e. DDPG, MADQN and MADDPG with both MLP and ATT schemes. Random access and simple Q-learning are taken as the baseline without any channels state fusion. In Adaptive Partner CSS, since one SU takes all the others as its partners, there is $N_{\text{coop}} \equiv 5$. In Reliable Partner CSS, although the number of partners could be 1 to 5, we still set $N_{\text{coop}} = 5$ for comparison fairness.

- Random access loses the sight of any prior channels information, so it has the lowest one-time success probability which can not be improved by training episodes.
- Q-learning also performs poorly in this scenario (actually it works well when there are only 2 SUs.). Since the lack of global observation, channels state is time-varying and unreliable which makes the environment non-stationary. As the number of SUs increasing, the fast-growing size of Q-table makes the training difficult to ergodic all possible status, and SUs are prevented to use the past experience to replay in straight forward [18]. As a result, it is not suitable for such large-scale model.
- No matter Reliable Partner CSS or Adaptive Partner CSS, it can help the one-time success probability to increase along with training episodes and finally converge at a satisfying value by MADQN, DDPG and MADDPG, which proves again that DRL approaches are powerful to solve complex problems in CRNs. However, since MADQN is a value-based learning method like Q-learning, even equipped with the same partners selection, it is defeated by the policy-based DDPG and MADDPG which are able to find the action in a continuous space.

TABLE I
EXISTING LITERATURES

Reference	Sharing Scheme	Optimization Objective	Multi-channel	Channel Fusion	DRL			Other
					Value-Based	Policy-based	Multi-agent	
[3]	Overlaying	Throughput	✗	✗			✗	Convex optimization
[4]	Underlaying	Throughput	✗	✗			✗	Convex optimization
[5]	Underlaying	Throughput/Accuracy	✓	✗	Multi-DQN		✓	
[6]	Overlaying	Accuracy	✓	✓	Multi-DQN		✓	
[7]	Overlaying	Detected Energy [†]	✓	✗	DQN		✗	
[8]	Underlaying	Throughput	✗	✓			✓	Sequential analysis
[9]	Overlaying	Throughput/Accuracy	✓	✗	Q-learning		✗	
[10]	Not mentioned	Accuracy	✓	✓ [‡]	Multi-DQN		✓	
[11]	RACH	Accuracy	✗	✗	Multi Q-learning		✓	
[12]	Overlaying	Accuracy	✓	✗	DQN		✗	
[13]	Underlaying	Throughput	✓	✗	DQN and Game		✗	
[14]	Not mentioned	Accuracy	✓	✓	DQN		✗	
[15]	Overlaying	Detected Energy	✓	✓	CNN		✗	
[16]	Not mentioned	Throughput	✓	✗		DDPG	✗	
[17]	Overlaying	Throughput	✓	✗	Multi-DQN		✗	
Our work	Overlaying	Accuracy	✓	✓		MADDPG	✓	

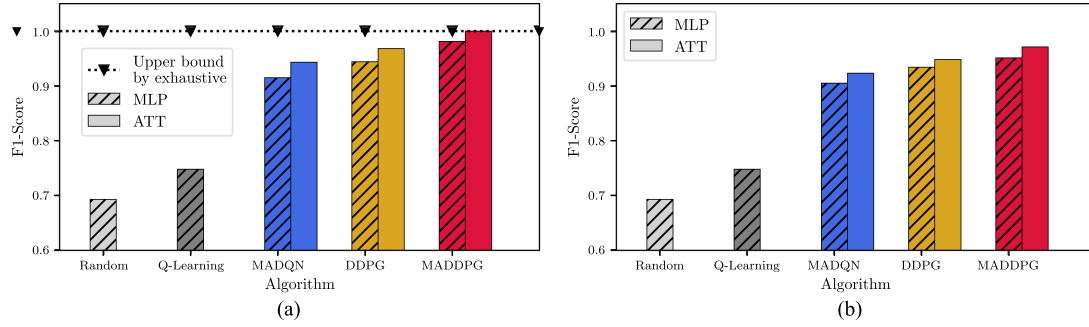
[†] Aim to minimize the average detected energy. The lower the better.[‡] Taking the decay of channels belief into consideration.

Fig. 6. F1 scores with different algorithms. (a) F1-Score in fixed pattern. (b) F1-Score in dynamic access pattern.

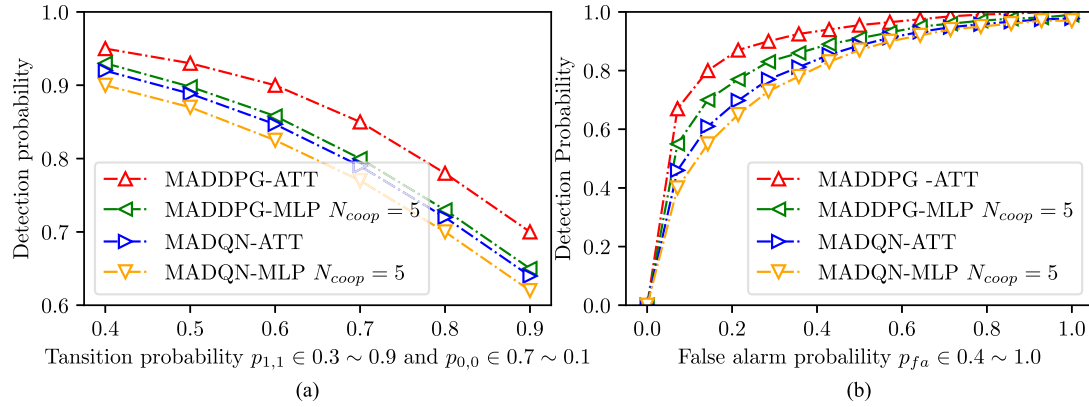


Fig. 7. Impact of Markov transition and false alarm on one-time success probability. (a) Different Transition Probability (b) Region of Convergence.

- With the same DRL algorithm, the one-time success probability can be further improved by Adaptive Partner CSS compared with Reliable Partner CSS. That is because channels state sensed by a reliable partner may out of reference due to the channels fading and shadowing. While Adaptive Partner CSS can be operated with the comprehensive consideration of both the reliability and geographical distribution of SUs. Such phenomenon is further interpreted in Section VI-D.

Shown as Fig. 6, two scenarios are considered to calculate F1 scores of all the algorithms mentioned above, i.e., the fixed pattern in which all channels take turns to be activated following a fixed pattern [12], and the dynamic access pattern adopted in the paper. Since the exhaustive algorithm is out of work when the channels state is unknown, it is only taken as the upper bound for F1 scores in the fixed pattern.

- In the fixed pattern, there exists the optimal solution [12] that can be obtained by the exhaustive algorithm, which is

TABLE II
MADDPG NEURAL NETWORK CONFIGURATION

Name	Neurons Num. and Active Fun.	Type
Input layer	$u_{a,0} = N_{fb}$ for s_n , ReLU	Actor
Hidden Layer	2 layers with $u_{a,1} = u_{a,2} = 300$, ReLU6	
Output layer	$u_{a,3} = 1$ for a_n , Sigmoid	
Input layer	$u_{c,0} = N_{fb} \times N + N^\dagger$ for (S, A) , ReLU	Critic-MLP
Hidden Layer	2 layers with $u_{c,1} = u_{c,2} = 300$, ReLU6	
Output layer	$u_{c,3} = 1^\ddagger$ for $\nabla_\mu J(\mu)$, NA	
Input layer	$N_{fb} \times N + N^\dagger$ for (S, A) , ReLU	Critic-ATT
Attention Layer*	$N_{head} = 4$ attention heads, ReLU6	
Output layer	1 for $\nabla_\mu J(\mu)^\ddagger$, NA	

† The inputs of critic are the global state-action pairs (S, A) of N agents with the dimension of $(N_{fb} + 1) \times N$.

‡ The output of critic is the policy gradient value scalar with the dimension of 1.

* The attention layer is embedded in MADDPG-ATT, and the multi-head attention networks is detailed in [29].

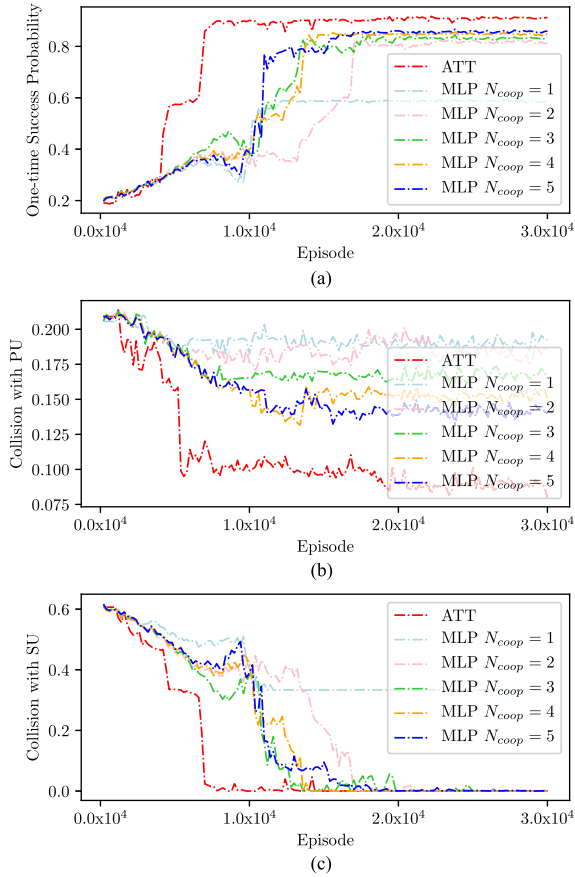


Fig. 8. Impact of cooperative partners number on one-time success probability. (a) Channel Selection Accuracy. (b) SUs-PUs Collision. (c) SUs Collision.

taken as the upper bound for F1 scores comparison. In the dynamic access pattern, F1 scores is deteriorated because the channel state is time-varying and the probability of state transition is indeterminate. The exhaustive algorithm is even out of work in this condition. However, MADDPG-ATT still performs well compared with others, which is almost close to 1. Since all the algorithms reveal similar performance compared with that in Fig. 5, we only take the one-time success probability for the following performance comparison unless otherwise stated.

- As a centralized algorithm, the actor network in DDPG needs the concatenation state of all agents to construct the global state. Hence, the communication overhead greatly blooms as the number of SUs increases. Since the paper mainly focuses on the distributive algorithms, for easy-to-access, we take MADDPG and MADQN for the following performance comparison.

B. Impact of Transition Probability and False Alarm

When PU_m transmits with the pre-fixed transmission power, spectrum sensing based on the energy detection at SU_n is further affected by $h_{m,n}$, which dynamically changes with the noise, channels shadowing and multi-path fading effect. Besides, channels occupation of PUs varies from time to time, but SUs can only sense parts of the overall frequency bands once a time. Thus, such energy detection is not always reliable, i.e., the results sensed by SUs may not be exactly the same with the ground truth of channels occupation.

Generally, there is the probability of false alarm p_{fa} and miss detection p_{md} for channels state. In specific, when SU_n falsely senses a vacant channel ($o_f = 1$) as occupied ($u_{n,f} > \Gamma_n$), such error is known as *false alarm*. Correspondingly, when SU_n misses a channel occupied by PUs ($o_f = 0$) as a vacant channel ($u_{n,f} < \Gamma_n$), it is known as *miss detection*.

- Fig. 7(a) discusseses the impact of primary channels' Markov transition probability on the sensing accuracy. When $p_{0,0}$ changes from 0.3 to 0.9 (i.e., channels gradually incline to be occupied by PUs.) and the vacant probability $p_{1,1}$ shrinks from 0.7 to 0.1, the one-time success probability declines as channels becoming busy (e.g., from 95% to approximately 70% in MADDPG-ATT). However, MADDPG-ATT still presents the advantage in finding a vacant channel by at least 8%, 6% and 5% performance improvement compared with MADQN-MLP, MADQN-ATT and MADDPG-MLP, respectively.
- Fig. 7(b) takes the average receiver operating characteristic (ROC) curves [7], [15] to reveal the algorithms' performance on treating the channels detection as a classification problem. Although all algorithms show comparable one-time success probability with a high value of p_{fa} , MADDPG has a significant advantage when p_{fa} is low, especially combined with Adaptive Partner CSS. In specific, the area under ROC curve (AUC) is calculated by 86.4%, 81.7%, 78.4% and 76.1% for MADDPG-ATT, MADDPG-MLP, MADQN-ATT and MADQN-MLP, respectively. Since MADDPG still performs the superiority for each SU working as a classifier, it is operated for the following performance comparison.

C. Impact of Cooperative Partners Number

Once SU_n initializes a data transmission, it first seeks a vacant channel with the largest probability according to the sensing strategy. There exist three conditions for channels collision: 1) The channel sensed by SU_n is neither occupied by any PUs or other SUs currently. Therefore, SU_n can access the channel successfully, which is known as the *one-time success*

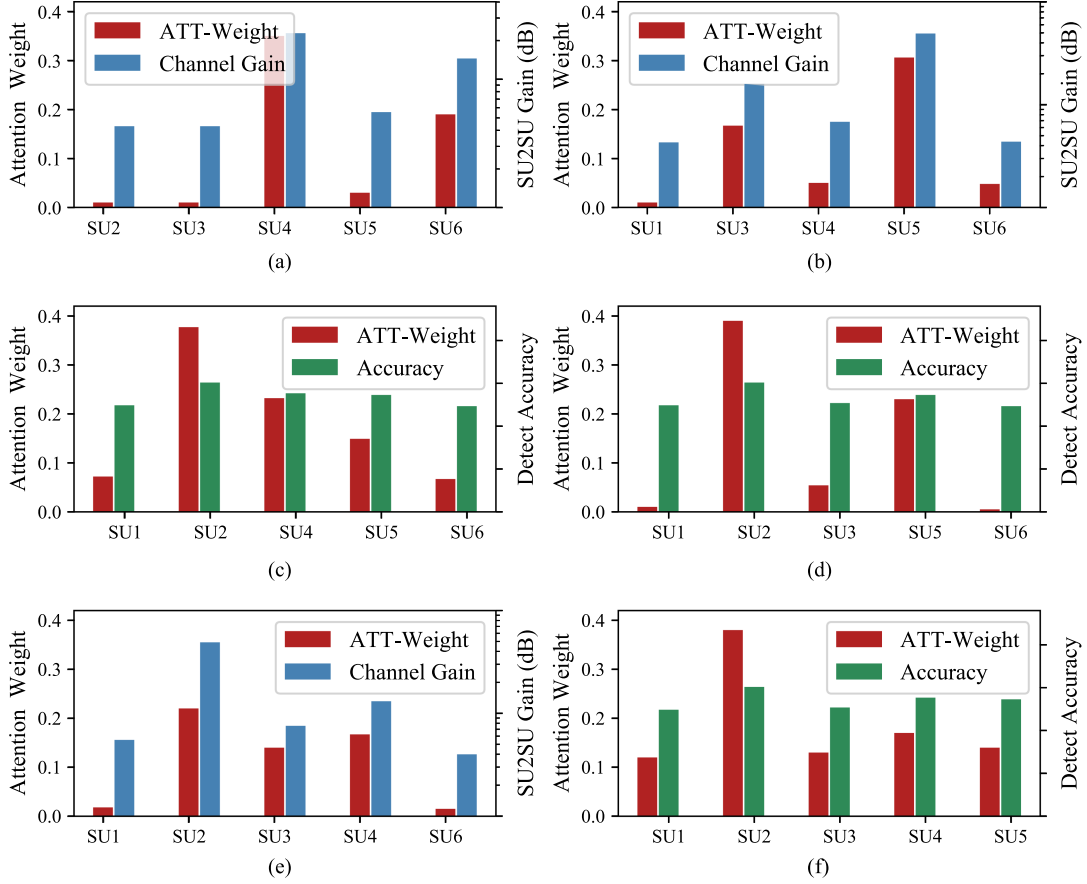


Fig. 9. Visualization of attention weight with the sensing accuracy and SUs-to-SUs channels gain in MADDPG-ATT. (a) Impact of Channel Gain to ATT-Weight at Agent 1. (b) Impact of Channel Gain to ATT-Weight at Agent 2. (c) Impact of Detect Accuracy to ATT-Weight at Agent 3. (d) Impact of Detect Accuracy to ATT-Weight at Agent 4. (e) Impact of Channel Gain to ATT-Weight at Agent 5. (f) Impact of Detect Accuracy to ATT-Weight at Agent 6.

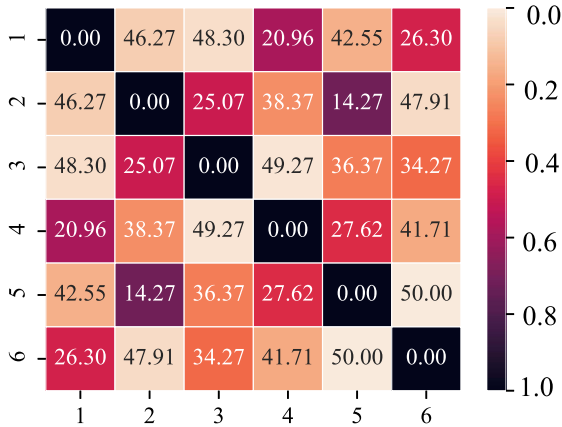


Fig. 10. Distance matrix and attention weight of SUs.

probability. 2) The channel sensed by SU_n is currently occupied by PUs, which leads to *SUs-PUs collision*. 3) When more than two SUs sense and access a vacant channel concurrently, *SUs collision* may happen. Therefore, the objective of CSS should not only enhance the one-time success probability, but also try to avoid the channels collisions.

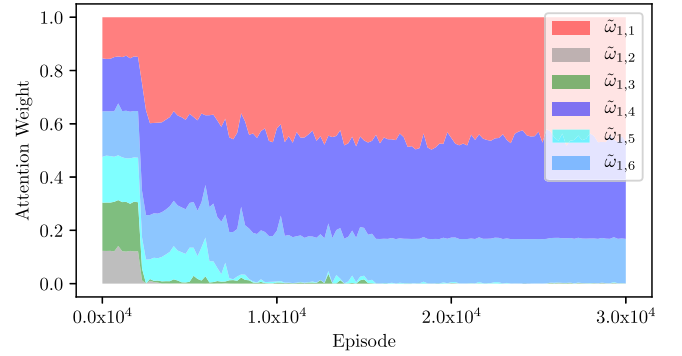


Fig. 11. Convergence of attention weight $\tilde{W}_1$ by MADDPG-ATT. (Although there is little fluctuation at the beginning, the attention weight of SUs finally converges at nearly 8000 episodes which is accordance with the results of Fig. 8. Because the convergence curves are similar for all SUs, we only take SU_1 for example.)

Fig. 8 illustrates the one-time success probability, SUs-PUs collision as well as SUs collision for MADDPG-MLP and MADDPG-ATT respectively. Since there are $N = 6$ SUs involved into the system, the number of partners for MADDPG-MLP N_{coop} increases from 1 to 5.

- For MADDPG-MLP, the one-time success probability blooms while the SUs-PUs collision and SUs collision reduces when more partners are involved into $\mathcal{G}_n$. It can be easily interpreted that when more partners participate into the channels state fusion, channels belief can be more reliable. In specific, each agent can obtain the recent channels state from its partners and fuse such state into its own belief by (12). Therefore, the actor network enables to obtain more abundant global state to improve the success accuracy when there are more partners.
- Compared with MADDPG-MLP when $N_{\text{coop}} = 5$, both the sensing accuracy and the channels collision can be further improved by MADDPG-ATT. The results reveal the limitation of Reliable Partner CSS which only takes the reliability into consideration but ignores the geographical distribution of SUs.

D. Visualization and Interpretation by Attention Weight.

It should be noticed that according to (23), the reward of MADDPG-ATT is actually the negative value of detected energy at a specific channel, which implicates the channels fading between PUs and SUs. Since PUs always transmit with a fixed power, the distance between SUs related to SUs-to-SUs channels gain can be indirectly learned by agents.

Fig. 9 reveals the impact of sensing accuracy (i.e., the reliability in statistic) and SUs-to-SUs channels gain (i.e., SUs-to-SUs distance) on the attention weight when the networks have been converged. The distance matrix among SUs is depicted in Fig. 10, where the depth of shadow is consistent with the SUs-to-SUs distance.

- When the sensing accuracy of all partners is nearly the same (from more than 80% to nearly 85%, which is omitted in the figures for clarity), the distance related SUs-to-SUs channels gain is the dominate factor for partner selection. Taking SU_1 for example, its attention weight converges along with the training iterations in Fig. 11 and the converged values are in order of the SUs-to-SUs channels gain $h_{n,i} (1 \leq i \leq N)^4$. The attention weights of the most nearby partners, i.e., SU_4 and SU_6 to SU_1 have the largest values, which is in consistence with the distance matrix shown in Fig. 10. Similar conclusions can be obtained for SU_2 and SU_5 in Fig. 9(b) and (e), respectively.
- On the contrary, for SU_3 , SU_4 and SU_6 in Fig. 9(c), (d) and (f) respectively, the attention weight is more related to the sensing accuracy. Take SU_3 for example. SU_4 , which is farthest from SU_3 , has a larger attention weight than SU_6 who is more close to SU_3 . Now the reliability of SUs becomes the dominant factor for partner selection.

Actually, SUs' reliability and channels gain are inter-involved in Adaptive Partner CSS. When one factor is dominant, it does not mean the other factor doesn't play a role. Such cases above can interpret that the comprehensive consideration of reliability and geographical distribution of SUs by MADDPG-ATT does bring further performance improvement compared with Reliable Partner CSS.

⁴ $h_{n,i} \propto d_{n,i}^{-q}$ (according to the free-space path loss model, $q = 2$), where $d_{n,i}$ is the distance between SU_n and SU_i .

VII. CONCLUSION

Based on the proposed Reliable Partner CSS and Adaptive Partner CSS schemes, SUs can cooperatively sense a vacant channel for opportunistic transmission in CRNs. However, the network topology, channels fading and the occupation pattern of PUs are difficult to be predicted. Hence DRL approaches have been adopted to help mapping the high-dimension solution space to an optimal action. Thanks to the 'centralized training and decentralized execution' feature of MADDPG, each SU is endowed with the global awareness of channels state by its partial observation, which greatly reduces the synchronization and communication overhead caused by the cooperative spectrum sensing.

Multi-head attention mechanism is further introduced to reveal the inner-connection of SUs. By the visualization of attention weight, it can be interpreted that compared with Reliable Partner CSS, Adaptive Partner CSS is more reasonable and explicable because of the comprehensive consideration of reliability and geographical distribution of SUs.

Numerical simulations demonstrate that by the proposed DRL based CSS algorithms, the sensing accuracy can be greatly enhanced comparing with other non-cooperative learning methods. In specific, compared with centralized or value-based learning, MADDPG has the strengths in finding the optimum action in a continuous space and reducing the communication overhead. While compared with Reliable Partner CSS, Adaptive Partner CSS can further enhance the system performance which can be interpreted by the visualization of attention weight.

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